DFT-Validated Evidence of Ferromagnetic Mislabeling Among Dzyaloshinskii–Moriya-Candidate Materials in the Materials Project Database

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High-throughput databases such as the Materials Project (MP) are widely used as the first filter in computational searches for Dzyaloshinskii–Moriya-interaction (DMI) hosts for chiral spin-texture and skyrmion-based non-volatile memory. These searches typically trust the database's recorded magnetic ordering, even though MP's automated relaxation workflow initializes every calculation from a single ferromagnetic (FM) moment guess and rarely explores antiferromagnetic (AFM) alternatives. We report a direct density-functional-theory (DFT) test of this assumption on a structural family – noncentrosymmetric, heavy-spin-orbit-coupling-element, net-moment compounds – selected because it is exactly the population that DMI-candidate screening pipelines draw from. Using MP's own energy-mapping protocol, with documented deviations only where it fails to converge, we enumerate and converge FM and symmetry-distinct AFM orderings for 26 candidates drawn from a machine-learning-filtered shortlist built on a seven-source, 64,635-record harmonized materials database; every calculation passes a per-ion spin-state audit or is re-run, and marginal cases are re-decided with tight same-cell FM/AFM pairs. Twenty-one of the 26 MP "ground-truth" FM labels are overturned by a lower-energy AFM state (margins of 0.19–24.68 meV/atom), three are confirmed FM, and two are degenerate to within electronic-convergence precision. This DFT-level result independently corroborates, in a materially different and more consequential population, the large-N statistical finding of systematic FM bias in MP recently reported from MAGNDATA-trained ML classifiers [Fahmy, arXiv:2509.05909]. For one confirmed-FM candidate, $Li_4Fe_2TeWO_{12}$ (mp-754090), we go one step further and extract both the isotropic exchange constant ($J = 2.78$ meV per Fe–Fe bond on the twelve-neighbor close-packed Fe sublattice, from a same-cell FM/AFM energy mapping) and one projection of the DMI vector ($D = 0.10$ meV, $D/J \approx 3.5\%$, from a constrained noncollinear+spin-orbit canting-angle probe) directly from DFT, and use the exchange, unmodified, as the input to finite-temperature LAMMPS-SPIN dynamics; for a second confirmed-FM candidate, Tb_6FeBi_2 (mp-1079285), constrained spin spirals resolve both symmetry-allowed components of the Fe-chain DMI vector ($|D| = 0.10$ meV per bond, $D/J \approx 0.7\%$). Starting from the ordered ground state, the Fe-sublattice magnetization of mp-754090 falls from $|m| = 0.97$ at 10 K to the finite-size floor above 125 K, and a Binder-cumulant crossing over four system sizes (256–4000 spins) places the ordering temperature of the model at 101 ± 1 K, matching the classical-Heisenberg fcc estimate (≈103 K) for the same J and falling 22% below mean field (≈129 K, which therefore overestimates it by 28%); the curve is reproduced by an independent Monte Carlo calculation. The three-shell model of mp-1079285 ($zJ = 52.6$ meV) gives a crossing at 139 ± 3 K. Taken together, these results argue that MP's ordering label should be treated as a screening prior rather than ground truth for exactly the structural family that DMI-material discovery pipelines target, and they demonstrate what a full, DFT-grounded verification of one candidate costs and returns.

I. INTRODUCTION

Materials that host a stable Dzyaloshinskii–Moriya interaction (DMI) [1, 2] are of active interest for chiral spin-texture and skyrmion-based non-volatile memory, because the antisymmetric exchange term the DMI contributes can stabilize spin spirals and skyrmion lattices at zero field [3–5]. Skyrmion lattices of this kind have been observed in bulk chiral magnets [6, 7], at atomic scale in ultrathin films [8], written and deleted individually with a spin-polarized current [9], and stabilized at room temperature in multilayer stacks [10]; the materials landscape and the device prospects that motivate this search are reviewed in Refs. [11, 12]. Searching for such materials computationally begins, almost universally, with a curated high-throughput database – most commonly the Materials Project (MP) [13] – used to filter tens of thousands of candidates down to a tractable shortlist before committing expensive first-principles or simulation resources. That filtering step depends on the database's recorded magnetic ordering (ferromagnetic, antiferromagnetic, ferrimagnetic, or non-magnetic) being correct, or at least an unbiased estimate.

It is well documented that automated high-throughput DFT workflows are prone to a specific failure mode here: relaxing a structure from a single initial magnetic-moment guess – almost always ferromagnetic – converges to whichever local minimum that guess is closest to, without necessarily finding the true magnetic ground state [14]. Materials Project's own magnetic-ordering methodology paper is explicit that a full energy-mapping analysis, enumerating symmetry-distinct antiferromag-

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netic configurations and comparing converged energies, is required to determine the ground state reliably [14]; the bulk of MP's own entries, generated at scale, do not receive this treatment.

Recent independent evidence, obtained through a very different route, points at the same problem. Fahmy [15] trained propagation-vector and magnetic-order classifiers on the MAGNDATA database of experimentally solved magnetic structures [16, 17] and applied them to the Materials Project database, finding a systematic disagreement consistent with a ferromagnetic bias affecting more than 7,843 MP-labeled compounds. That is a large-N, purely statistical detection: the classifier disagrees with the label, but no first-principles calculation is performed on any individual compound to confirm which side is right.

This work asks a complementary question at higher fidelity but lower N: within the specific structural family that DMI-candidate screening pipelines actually select – noncentrosymmetric space groups, a net magnetic moment, and a heavy spin-orbit-coupling element present – how often is MP's stored FM label the true DFT ground state? We build a seven-source harmonized materials database and a bias-aware magnetic-ordering classifier, use them to filter for DMI-compatible candidates, and then run the same collinear energy-mapping protocol MP itself prescribes [14] on 26 of the resulting short-listed candidates. The result is a direct, per-compound DFT verdict rather than a statistical classifier disagreement, obtained specifically in the population most consequential for DMI-material discovery. We then carry two confirmed-FM candidates, $Li_4Fe_2TeWO_{12}$ (mp-754090) and Tb_6FeBi_2 (mp-1079285), through a full pipeline: DFT extraction of the isotropic exchange constants and of the DMI, and finite-temperature atomistic spin dynamics driven by those DFT-derived constants with no fitting or rescaling, with the ordering temperature of each model fixed by finite-size scaling. For the third confirmed-FM candidate, $FeP_6(WO_8)_3$ (mp-850225), we resolve its two exchange shells.

II. METHODS

A. Harmonized multi-source database

Seven public sources – Materials Project [13] (37,783 records, via the `mp-api` REST interface [18]), JARVIS-DFT [19] (19,116), AFLOW [20] (2,639), MAGNDATA [16, 17] (1,919 of 2,414 attempted), 2DMatPedia [21] (1,862), OQMD [22, 23] (816), and NOMAD [24] (500) – were merged into one schema (`material_id`, `formula`, `elements`, `space_group`, `density`, `volume`, `band_gap`, `ordering`, `total_magnetization`, and, where available, full structure and per-site magnetic-moment vectors), for 64,635 total records. Records retain a source tag rather than being silently deduplicated across sources. MAGNDATA is the only source with experimentally determined (rather than DFT-derived) magnetic structure, and is accordingly the ground-truth source for classifier training and validation. A new end-to-end scraper against the live Bilbao Crystallographic Server index recovered 1,919 of 2,414 entries (79.5%). The 495 failures break down as 449 HTTP 404s from the server's temporary `.mcif` endpoint, 41 structures the parser refuses (36 explicitly disordered, 5 incommensurate), 3 read timeouts, and 2 entries whose fetched CIF contained no structure. Within the 404s, 148 are entries whose formula carries fractional site occupancies and 3 carry HTML markup in the upstream formula string, which makes the request URL malformed; those 151, together with the 41 refusals, are the at most 192 entries attributable to a data-format limitation. The remaining 298 are 404s on well-formed URLs for integer-stoichiometry entries, and they are transient rather than permanent: they persist across the scraper's four in-run retries, but re-requesting one of them (entry 0.57, $ScFeO_3$) returned 404 and then 200 within minutes, while its entry page is served throughout. 79.5% is therefore a property of this run rather than a ceiling, and a slower retry campaign should recover most of the 298. Nothing below depends on the missing entries, all of which are simply absent from the training and validation sets, but the MAGNDATA ground truth used here is smaller than the source allows. Ordering was validated on a known case ($LaMnO_3$, correctly recovered as AFM, space group *Pnma*).

B. Structure-aware magnetic-ordering classifier

A LightGBM [25] gradient-boosted-decision-tree classifier predicts ordering (FM/AFM/FiM/NM) from composition- and structure-derived features, in the spirit of general-purpose composition-based property models [26, 27] and of earlier machine-learning treatments of magnetic order and ordering temperature [28–31]. A composition-only baseline (per-element statistics of atomic number, mass, electronegativity, atomic radius, plus group/period/transition-metal/rare-earth fractions) achieves 56.7% accuracy (60.7% balanced accuracy). Adding two tiers of structural features – broad, near-universal-coverage fields (space group number, crystal-system bucket, unit-cell volume, density, site count) and local-geometry features around magnetic sites (nearest magnetic–magnetic neighbor distance, magnetic coordination number, fraction of magnetic sites; computed from atomic positions and species only, never from the magnetic-moment vectors that generate the label itself) – raises this to 67.4% accuracy (68.0% balanced accuracy) on a stratified 28,769/7,193 train/test split (Fig. 1). In the first version of this pipeline the local-geometry tier was computed only for MAGNDATA, whose CIFs carry atomic positions inline; every other source was represented by its metadata alone. Because MAGNDATA is the one source whose labels are experimental and 83%

AFM, a "Tier-2 present" pattern then acted as a proxy for the source database rather than for geometry. We therefore fetched structures for every source with a public structure endpoint (Materials Project 37,783/37,783, JARVIS 19,116/19,116, AFLOW 2,637/2,639, NOMAD 495/500, 2DMatPedia 1,862/1,862; OQMD 0/816 because its REST endpoint was unreachable throughout this revision) and recomputed the tier for all of them. The change leaves the headline numbers essentially unchanged (67.0% → 67.4% accuracy, 68.1% → 68.0% balanced) but moves the model's reliance where it belongs: per-source test accuracy on Materials Project rises from 65.6% to 66.9% (balanced 63.0% → 65.9%), while the inflated MAGNDATA figure drops from 89.5% to 75.8%, and four local-geometry features (mean/std/min magnetic–magnetic distance, mean magnetic coordination) enter the top seven features by LightGBM split importance, where before none appeared in the top ten (Fig. 1b). Space-group number remains the single most important feature, despite being absent from the baseline entirely. This classifier is retrained on all labeled data and deployed to predict ordering for every record lacking a ground-truth label, including all 19,116 JARVIS records, none of which carry a populated ordering field.

C. DMI-compatibility screening funnel

The full 64,635-record database, with ordering backfilled by the deployment classifier where no ground-truth label exists, is filtered for candidates plausibly compatible with a DMI: a net magnetic moment (FM or FiM; AFM is excluded, since a texture-hosting net moment is a hard requirement for the downstream simulation stage), a noncentrosymmetric space group (DMI is symmetry-forbidden under an inversion center [2]; verified computationally against all 230 space groups, recovering the textbook 92 centrosymmetric / 138 noncentrosymmetric split), presence of at least one heavy spin-orbit-coupling element (Ta, W, Re, Os, Ir, Pt, Au, Hg, Bi – the classic DMI-enhancing species by the Fert–Levy and Moriya mechanisms [2, 32], whose interfacial and bulk manifestations have been characterized both experimentally and from first principles [33, 34]), and thermodynamic plausibility ($E_{\mathrm{hull}} \leq 0.1$ eV/atom where known). Of 64,635 records, 47,614 carry a net-moment ordering, 19,280 of those are noncentrosymmetric, and 1,864 also contain a heavy-SOC element (Materials Project 768, JARVIS 537, AFLOW 392, 2DMatPedia 77, NOMAD 53, OQMD 35, MAGNDATA 2; Fig. 2). A hull energy is known for 1,340 of them (the MP, JARVIS and OQMD records), 671 of which lie within 0.1 eV/atom of it. Two revisions to this step were made during the present audit. First, the `ehull` field of the JARVIS-DFT snapshot used here is not a hull distance (its median over the 93,902-entry snapshot is 1.7 eV/atom, the known stable compound TbMnSi carries `ehull`= 2.45 eV/atom, and 101 of the 3,206 harmonized JARVIS records with a *positive* formation energy carry `ehull`= 0 — among them $BaFeHg_4$ at +0.36 eV/atom, which is how the inconsistency was found); it was therefore discarded and the hull distance of every JARVIS record recomputed from JARVIS's own OptB88vdW formation energies with a per-chemical-system convex hull (13,967 of 19,116 records within 0.1 eV/atom). Three of the four JARVIS entries in the original top-30 shortlist fail this recomputed criterion (Mg_6FeBi 0.13, $BaFeHg_4$ 0.36, Ba_4ScHg 0.54 eV/atom); none of them entered the DFT batch, which is Materials Project only. Second, the hull criterion is now available as a hard filter: dropping every record with a *known* hull distance above 0.1 eV/atom (records from sources that provide no hull energy – AFLOW, NOMAD, 2DMatPedia, MAGNDATA – are retained rather than penalised for a missing field) leaves 1,195 candidates. All 26 candidates of the DFT batch survive the hard filter, and the original ranking from which that batch was drawn, in which the hull acted as a sort key rather than an exclusion, is kept alongside it so the batch remains reproducible; because the hard-filtered ranking orders stable records by hull distance first, the same 26 candidates spread from rank 1 to rank 663 under it, i.e. the DFT batch is not a sample of the funnel's most stable members (Fig. 3).

For the top 100 screened candidates per source, structures were backfilled via each source's own public data-access method (298 of 300 attempted backfills succeeded; two AFLOW records failed on repeated TLS errors from the AFLUX endpoint) and re-relaxed with a MACE foundation-model potential [35, 36], fine-tuned on this project's harmonized dataset via multihead training (a replay head sampling 20,000 configurations from the original pretraining distribution to avoid catastrophic forgetting), as an independent consistency check against each source's own DFT relaxation. The base model is MACE-MP-0 `small` (`num_channels`= 128, `max_L`= 1, two interaction layers, $r_{\max}$ = 5.0 Å) [36]; the relaxations use the Stage-2/SWA checkpoint of the fine-tune with its default head, and are static FIRE [37] optimizations in ASE [38] (no molecular dynamics anywhere in this work). Other universal potentials trained on Materials Project data, such as M3GNet [39] and CHGNet [40], would serve the same consistency-check purpose. All 298 relaxations converged (max force ≤ 0.05 eV/Å) with a median relaxation-energy shift of 0.0009 eV/atom relative to the source structure, indicating strong agreement between the fine-tuned potential and each source's own DFT. Classifier confidence and MACE relaxation-consistency were merged into a final ranking (converged → hull-stable → smallest relaxation shift → classifier confidence); the resulting top-30 shortlist is the pool this work's DFT validation batch is drawn from.

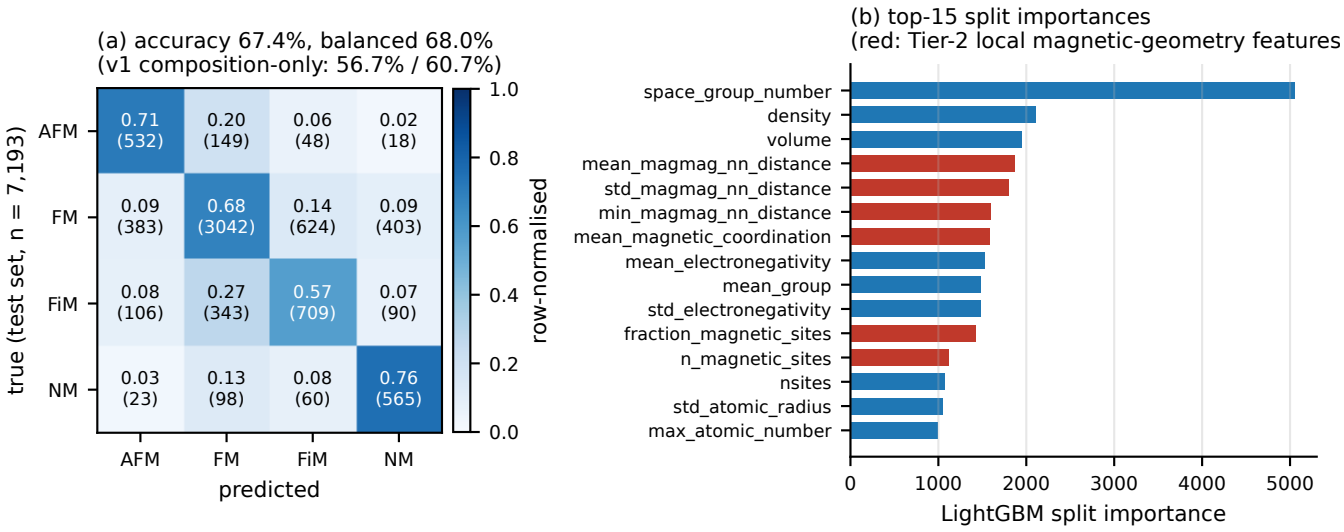


FIG. 1. Structure-aware ordering classifier on the held-out test set (7,193 records, stratified 80/20 split). (a) Row-normalised confusion matrix (counts in parentheses): the dominant confusion is FM → FiM and FiM → FM, i.e. between the two net-moment classes that the screening funnel treats identically, whereas AFM and NM are recovered at 71% and 76%. (b) The fifteen most-used features by LightGBM split count; the six Tier-2 local-magnetic-geometry features (red), computed from atomic positions and species only, now sit alongside the space-group and density/volume descriptors instead of acting as a MAGNDATA source flag.

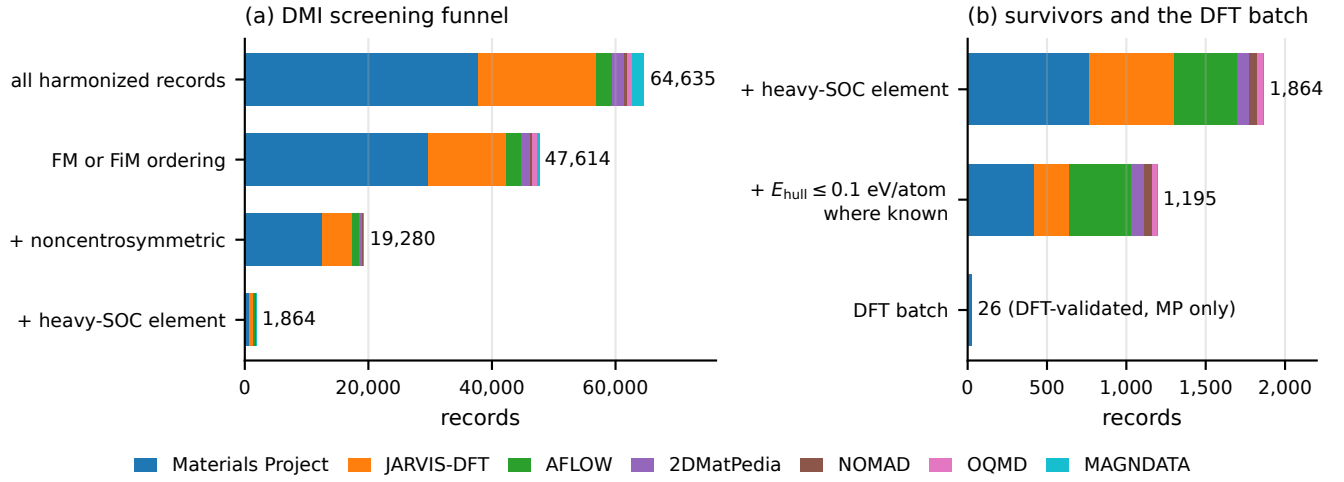


FIG. 2. The DMI-compatibility screening funnel by source database. (a) Each successive criterion – net-moment ordering (ground-truth label where present, otherwise the deployed classifier's prediction), noncentrosymmetric space group, and at least one heavy spin-orbit-coupling element – applied to the 64,635-record harmonized database. (b) The 1,864 survivors, the 1,195 that remain when a known hull distance above 0.1 eV/atom is used as a hard exclusion (sources without a hull field are retained), and the 26-candidate DFT batch, which is Materials Project only.

D. DFT ordering-validation protocol

For a subset of shortlisted candidates carrying an `ordering_source=ground_truth`, `confidence=1.0` label from Materials Project (i.e., MP's own recorded ordering, not a classifier prediction), we apply the same collinear magnetic-ordering energy-mapping approach behind MP's own workflow [14]: symmetry-distinct FM and AFM spin configurations are enumerated with `pymatgen.MagneticStructureEnumerator` [41] (backed by `enumlib`); VASP inputs are generated with `MPRelaxSet` – MP's own PBE exchange-correlation functional [42], GGA+U parameters [43–45] (e.g. LDAUU= 5.3 eV on Fe), plane-wave cutoff (520 eV), and Monkhorst–Pack k-point convention [46] – overriding only the initial MAGMOM per configuration; and

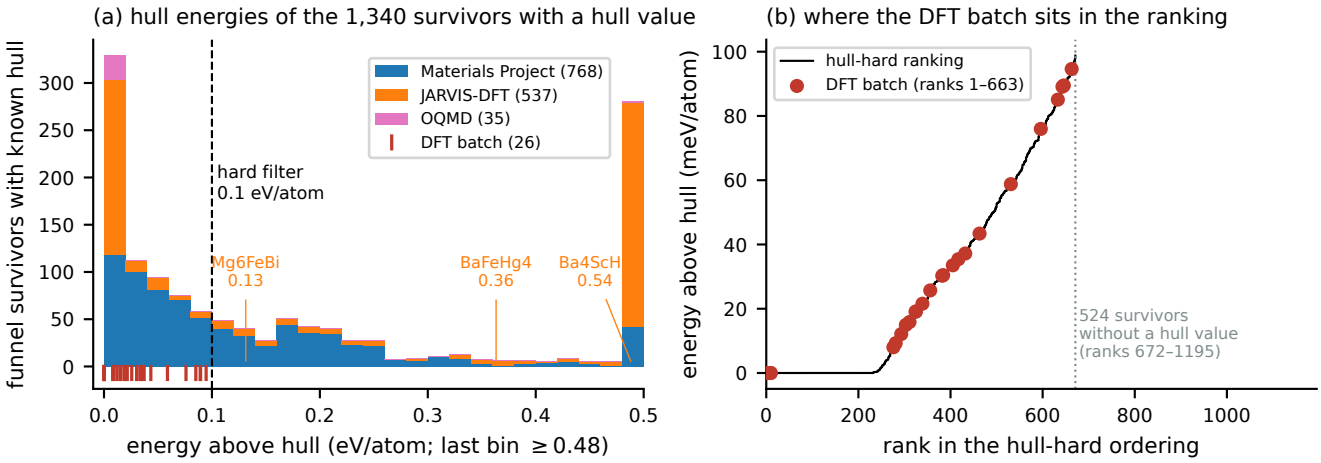


FIG. 3. Thermodynamic stability of the funnel survivors. (a) Distribution of the hull distance for the 1,340 survivors that carry one (Materials Project and OQMD as stored; JARVIS recomputed from its own formation energies, see text), with the 0.1 eV/atom hard filter and the three JARVIS entries of the original top-30 shortlist that it removes; red ticks mark the 26 DFT-batch candidates. (b) Hull distance versus rank in the hard-filtered ordering, which sorts stable records by hull distance first: the DFT batch (red) spans ranks 1–663, so it is a sample of the funnel's DMI-compatible population, not of its most stable members.

each configuration is relaxed to convergence within Kohn–Sham density-functional theory [47, 48] as implemented in VASP 6.4.3 [49] with the projector-augmented-wave method [50, 51] (`MPRelaxSet` defaults ISIF= 3, ISMEAR= −5, i.e. the tetrahedron method with Blöchl corrections [52], EDIFF= 5×10^{-5} eV per atom, i.e. 0.05 meV/atom). This choice keeps the resulting energies directly comparable to MP's own stored values rather than constituting an arbitrary independent DFT recipe. Beyond VASP's own convergence flag, every calculation was audited in two further ways: the final RMS charge-density residual was checked (the "EDIFF is reached" banner can fire on a transient while the density is still far from self-consistent), and the final per-ion moments were compared against the intended MAGMOM pattern and against the moment magnitudes of the FM calculation of the same compound. The second check matters: a configuration whose magnetic ions collapse to a low-spin or nonmagnetic state, or whose moments re-order into the FM pattern during the SCF cycle (a doubled "AFM" cell that is really FM), passes VASP's convergence test but is not a sample of the ordering it was labeled with, and is excluded from the verdict rather than counted.

The first pass (93 relaxations across the 26 candidates; 116 audited relaxations in total once the second batch below is included) left 12 calculations in seven candidates that failed the spin-state audit: collapsed or low-spin Fe/Cr moments in an AFM configuration (mp-1173139, mp-1173143, mp-551086, mp-775194), "AFM" cells whose moments re-ordered to the FM pattern during the SCF cycle (mp-682554), an "AFM" pattern that had flipped only a nonmagnetic d^0 Ti site and was therefore FM in disguise (mp-1216981, whose apparent zero margin in an earlier draft was an artefact of exactly this), and an FM reference that collapsed to a nonmagnetic state (mp-22972). Rather than discarding these candidates, a second batch replaced every failed configuration with a freshly enumerated one in a spin state that the audit accepts: additional symmetry-distinct AFM/FiM cells for mp-1173139, mp-1173143, mp-775194 and mp-551086 (the ferrimagnetic ones with the two Fe sublattices, or Fe and Cr, unequal in number), Fe-only AFM supercells for mp-1216981 (Ti excluded from the enumeration), anti-collapse FM reruns for mp-22972 (in the primitive cell and in the cell of its lowest AFM configuration), and, for mp-682554, whose $1 \times 1 \times 2$ Fe–Fe stacking (130 atoms, 27.9 Å between the two Fe sites) diverged twice at the first density-mixing step through charge sloshing, five single-point mixer/smearing variants of the same cell run side by side, with the production relaxation restarted only under the first variant that converged. A superseded attempt (a diverged or collapsed run that was re-done) is neither a sample nor a gap in the count.

Two protocol effects turned out to matter at the sub-meV/atom level. First, the MP recipe compares an FM relaxation in the primitive cell with AFM relaxations in supercells, each with its own k-mesh and each relaxed with ISIF= 3 to its own cell shape; for the sixteen configurations run both ways this "loose" margin differs from a "tight" one (both orderings relaxed with ISIF= 2 in the same relaxed AFM cell, identical k-mesh, EDIFF= 10^{-6} eV, EDIFFG= −0.01 eV/Å) by up to 1.3 meV/atom in either direction ($Ba_4Ta_{10}FeO_{30}$: +1.18 → 0.00; $FeP_6(WO_8)_3$: +2.09 → +3.38). Every candidate whose verdict rested on a loose margin smaller than about 2 meV/atom was therefore re-decided with a

tight pair built in the cell of its lowest valid non-FM configuration (14 pairs in the first tight batch, 6 more for the remaining small-margin overturns), and a tight number replaces the loose one for that configuration wherever it exists. Second, the verdict rules are asymmetric by construction: a label is *overturned* as soon as one audited configuration lies more than 0.10 meV/atom below FM – a still-missing configuration cannot undo that – whereas *confirmed FM* and *degenerate* require every staged configuration to have a valid energy. The 0.10 meV/atom window is twice the per-atom EDIFF of the loose protocol.

All five tungsten-bearing candidates in this batch initially failed to converge under MP's standard Hubbard correction (LDAUU= 6.2 eV on W): tungsten in these oxides is typically W^{6+}, a nominally d^0 ion, and a large U applied to a near-empty d-shell leaves the self-consistent density with no restoring force, oscillating indefinitely rather than converging. Removing U from tungsten (LDAUL= −1) entirely, applied uniformly across every ordering of the affected compound, converged all 18 corresponding production calculations cleanly on the first retry. One of the five is not d^0: in $FeP_6(WO_8)_3$ (mp-850225), charge balance with Fe^{3+} and P^{5+} requires W^{5+} (d^1), and in the U-free runs each W carries 0.78–0.86 μ_B antiparallel to Fe in the FM state (Sec. III A). These reruns also fix the cell shape and volume at MP's relaxed values (ISIF= 2, ions only) and use Gaussian smearing (ISMEAR= 0, σ = 0.05 eV; tetrahedron smearing for mp-755203) for SCF stability. This changes the absolute energy scale for those five candidates relative to MP's own U-inclusive value, but not the internal FM-vs-AFM comparison within each compound, since the same settings are applied to every ordering of that compound identically.

E. Exchange and DMI extraction for mp-754090

For $Li_4Fe_2TeWO_{12}$ (mp-754090), confirmed FM by the protocol above, we additionally extract effective spin-Hamiltonian parameters directly from DFT by energy mapping, the standard route from the total energies of constructed spin configurations to Heisenberg parameters [53–55]. The isotropic nearest-shell exchange constant is obtained from the tight same-cell FM/AFM pair used for the ordering verdict (both orderings relaxed in the same cell, which is the fixed lattice a Heisenberg J is defined on). The Fe sublattice of the rock-salt-derived $Li_4Fe_2TeWO_{12}$ structure is close-packed: each of the two Fe sites (Fe0, Fe1) per formula-unit cell has twelve Fe neighbors, six in a triangular layer (5.112–5.157 Å, bond angles of 60°, 120° and 180°; four on the opposite sublattice and two on the same sublattice by lattice translation) and six across the 4.53 Å layer spacing (5.411–5.427 Å; again four on the opposite sublattice and two on the same). The next shell starts at 7.45 Å. For $H = -\sum_{\langle ij\rangle} J\,\hat{s}_i\cdot\hat{s}_j$ with each bond counted once, the collinear AFM configuration (Fe0 sublattice reversed) flips the eight inter-sublattice bonds per cell – four in-plane and four inter-layer – so $E_{\rm AFM} - E_{\rm FM} = 2\times 8\,J = 16J$; with the same-cell DFT value ΔE = 44.54 meV per two-Fe cell this gives $J = \Delta E/16$ = 2.78 meV per bond. The loose pair (each ordering at its own ISIF= 3 optimum) gives ΔE = 48.23 meV and J = 3.01 meV, an 8% protocol spread that we carry as the systematic uncertainty of J. (Two earlier versions of this analysis were wrong in ways worth recording: one quoted $J = \Delta E/4$, double-counting the flipped bonds; the next used a 5.25 Å neighbor cutoff that keeps only the six coplanar in-plane bonds and drops the six inter-layer ones, so that $\Delta E = 8J$ and J = 5.57 meV – the correct sum of an in-plane and an inter-layer bond, but on a lattice of *uncoupled* two-dimensional triangular layers, which has no long-range order at any finite temperature and gave a spurious $|m|(T)$ once the spin dynamics were properly equilibrated. The spin dynamics reported below use the twelve-neighbor model.) A single FM/AFM pair cannot separate the in-plane from the inter-layer exchange, nor resolve the four same-sublattice bonds, which are invisible to a two-configuration fit; all twelve bonds are assigned the same J in the spin model – an explicit assumption. This is a single-J effective model, not a multi-shell decomposition; its $zJ = 12J$ = 33.4 meV, which fixes the mean-field ordering temperature, is what the DFT pair determines directly.

DMI does not appear in a collinear FM-vs-AFM comparison at all; it requires spins canted relative to one another. We first checked whether mp-754090's crystal symmetry forces $D = 0$ via Moriya's rule (an inversion center at the bond midpoint is sufficient to forbid DMI exactly, at zero DFT cost) [2]: the specific Li/Fe/Te/W cation ordering in this structure reduces the space group to $P1$, with no symmetry operation beyond the identity, so no such shortcut applies and a real calculation is required. We then ran two static, constrained noncollinear-spin+spin-orbit-coupling calculations from the relaxed FM structure – the established route to a first-principles DMI [56, 57] – within unconstrained noncollinear PAW magnetism [58], using the penalty-energy method [59] (`I_CONSTRAINED_M=1`, constraining moment direction only) to hold Fe0 along $+\hat{z}$ and Fe1 along $+\hat{x}$ (canted$_+$) or $-\hat{x}$ (canted$_-$) – i.e. $\theta = \pm 90°$ canting in the xz-plane, an arbitrary but fixed reference axis since $P1$ symmetry does not privilege any particular plane. The moment constraint held essentially exactly in both calculations (Fe1 converged to $4.038\,\mu_B$ along x in canted$_+$ and $-4.039\,\mu_B$ in canted$_-$; Fe0 leakage off z was $< 0.002\,\mu_B$ in both). With the same bond-counting convention as for J ($z = 8$ inter-sublattice bonds per cell, each counted once; the four same-sublattice bonds connect parallel spins and contribute neither to the $\cos\theta$ nor to the $\sin\theta$ term), $E(\theta) = E_0 - zJ\cos\theta - zD\sin\theta$, and comparing $\theta = \pm 90°$ isolates the antisymmetric term

cleanly since $\cos(\pm 90^\circ) = 0$:

$$zD = \frac{E(\theta = -90^\circ) - E(\theta = +90^\circ)}{2}, \qquad (1)$$

giving $E(-90^\circ) - E(+90^\circ) = 1.546$ meV per cell and $D = 0.097$ meV per bond, i.e. $D/J \approx 3.5\%$. Because $\hat{s}_1 = \hat{z}$ and the canting is confined to the xz-plane, $\hat{s}_1 \times \hat{s}_2 = (0, \sin\theta, 0)$: this probe measures the mean y-component of the DMI vectors of the eight inequivalent inter-sublattice bonds in this lab frame specifically, not a generic magnitude, and should be read as a resolved DFT number for one projection of D rather than a fully characterized three-dimensional DMI vector. Since the ratio D/J is the physically meaningful quantity, it is unchanged by the bond-count correction above.

F. Exchange and DMI extraction for mp-1079285

The second confirmed ferromagnet, Tb_6FeBi_2 (mp-1079285, space group Pm, one Fe per primitive cell), is a different magnetic topology: the Fe atoms form chains along b (Fe–Fe 4.198 Å), the next Fe–Fe distances being 8.396 Å along $\pm 2b$ (chain next-nearest neighbours) and 8.396–8.407 Å to the six chain neighbours along $\pm a$, $\pm c$, $\pm(a + c)$. Tb is treated with the $4f$ electrons frozen in the core (`Tb_3` PAW), so the only moment in the model is Fe ($1.68\,\mu_B$ in the FM calculation), an explicit approximation that ignores any Tb–Fe exchange. Every collinear AFM configuration that the enumeration produced for this candidate turned out to be the same chain-antiparallel order in a different cell setting, so the tight pair in the $1 \times 2 \times 1$ cell resolves only the intra-chain coupling: with $H = -\sum_{\langle ij \rangle} J_{ij}\, \hat{s}_i \cdot \hat{s}_j$ (each bond counted once) it flips two chain bonds per cell, $E_{\mathrm{AFM}} - E_{\mathrm{FM}} = 4J_b$. A chain model alone has no order at any $T > 0$, so three further same-cell tight pairs were staged in supercells of the relaxed FM cell (`ISIF`= 2, `EDIFF`= 10^{-6} eV, `EDIFFG`= -0.01 eV/Å): the $2a$ and $2c$ cells with the two Fe antiparallel, which flip the $\pm a, \pm(a + c)$ and the $\pm c, \pm(a + c)$ bonds respectively ($\Delta E_{2a} = 4(J_a + J_{ac})$, $\Delta E_{2c} = 4(J_c + J_{ac})$), and a $1 \times 4 \times 1$ cell with the chain pattern $++--$, which flips two of the four chain bonds and all four $\pm 2b$ bonds ($\Delta E_{4b} = 4J_b + 8J_{2b}$). Two equations do not determine three inter-chain couplings; the model takes their mean, $J_\perp = (\Delta E_{2a} + \Delta E_{2c})/16$, under the stated assumption $J_a \approx J_c \approx J_{ac}$ (the three bond lengths agree to 0.01 Å). The $\pm 2b$ bond is resolved separately because its length coincides with the inter-chain shell, which no radial exchange function in the spin-dynamics code can separate (Sec. II G).

The DMI vector of the chain bond is constrained by symmetry: the Pm mirror plane is normal to b and contains the Fe site, so by Moriya's rules $\mathbf{D} \perp b$ and two projections determine $|\mathbf{D}|$. Each was obtained from a pair of 90° spin spirals in the $1 \times 4 \times 1$ cell (four Fe, moments constrained with `I_CONSTRAINED_M`= 1, $\lambda = 10$, spin-orbit coupling on, `EDIFF`= 10^{-7} eV, single points on the relaxed FM structure), rotating in the plane normal to $\hat{n}_1 \parallel a$ or $\hat{n}_2 = \hat{b} \times \hat{n}_1$ with either chirality. For a 90° spiral $\hat{s}_i \times \hat{s}_j = \pm\hat{n}$ on every chain bond while the symmetric exchange (including J_{2b}, whose bonds are antiparallel in both chiralities) is identical for the two, so $\mathbf{D}\hat{n} = [E(+) - E(-)]/8$ per bond. The constraint penalty energies were $\leq 1.6 \times 10^{-9}$ eV, the Fe moments had no component off their target axes (to the printed $0.001\,\mu_B$) with $|m_{\mathrm{Fe}}| = 1.50 \pm 0.01\,\mu_B$ in all four calculations, and the induced Tb moments ($\leq 0.123\,\mu_B$; Bi $\leq 0.002\,\mu_B$) rotate with the spiral, their magnitudes agreeing between chiralities to $0.001\,\mu_B$.

G. Finite-temperature spin dynamics

The DFT-derived $J = 2.78$ meV, with no fitting or rescaling, parameterizes a twelve-neighbor Heisenberg Hamiltonian for the Fe sublattice of mp-754090 (framework Li/Te/W/O sites carry no spin and are omitted from the geometry), implemented in LAMMPS-SPIN [60] within LAMMPS [61, 62] via the `spin/exchange` pair style (which counts each bond once, matching the convention of the DFT extraction) with a 5.5 Å cutoff that captures all twelve Fe–Fe bonds per site. `spin/exchange` uses the Bethe–Slater radial form $J(r) = 4a\,(r/d)^2 e^{-(r/d)^2}$ (with its b parameter set to zero); $d = 5.274$ Å is solved so that $J(5.130\text{ Å}) = J(5.418\text{ Å})$ exactly and a set so that both equal the DFT J, which makes $J(r)$ constant to within 0.06% over the two shells. A zero-step energy evaluation of the FM and Fe0-reversed configurations of the model reproduces the DFT $E_{\mathrm{AFM}} - E_{\mathrm{FM}} = 16J = 44.54$ meV per cell to 0.003%. The DMI is not included in the production sweep, for a reason that is itself a result: the LAMMPS `spin/dmi` pair style implements $\mathbf{D}_{ij} = D\,(\hat{e}_{ij} \times \hat{v})$ with a single global vector $\hat{v}$, and for the DFT probe geometry ($\hat{s}_i \times \hat{s}_j = \pm\hat{y}$) its energy difference is $D\,(\hat{v} \times \hat{y}) \cdot \sum_b \hat{e}_b$ over the eight inter-sublattice bond directions – a sum that is 0.03 (in units of a bond vector) for this near-centrosymmetric neighbor shell, so no choice of $\hat{v}$ can reproduce the DFT $E(-90^\circ) - E(+90^\circ)$ (a zero-step evaluation with $D = 0.097$ meV and $\hat{v} = \hat{y}$ gives 0.000 meV); the eight bond-resolved DMI vectors that a faithful model needs would require at least eight independent canting probes, which are outside the scope of this work (Sec. IV C). We instead bound the effect of a DMI of the measured magnitude on $|m|(T)$ with a control sweep that adds `spin/dmi` with $D = 0.097$ meV and $\hat{v} = \hat{y}$ (an arbitrary but representative placement of a 3.5% antisymmetric term) at four temperatures, and report the difference. The model is run on a $16 \times 8 \times 8$ replication (2048 spins) of the two-atom Fe primitive cell, with periodic boundaries and a rigid (spin-only) lattice. Single-ion anisotropy is set to zero (Sec. IV C). Each run starts from the fully aligned FM ground state and is evolved under Langevin spin dynamics ($dt = 0.1$ fs) – stochastic Landau–Lifshitz–Gilbert dynamics of clas-

sical spins [63, 64], the standard atomistic approach to finite-temperature magnetism [65, 66] – for 200,000 steps (20 ps) at twelve temperatures from 10 to 400 K; the normalized net magnetization $|m| = |\sum_i \hat{s}_i|/N$ is averaged over the second half of each run and over three independent thermostat seeds per temperature. The Langevin damping is set to $\alpha = 1$: the equilibrium $|m|(T)$ does not depend on α, but the thermalization time of the softest magnons of the finite cell scales as $1/\alpha$, and at the conventional $\alpha = 0.05$ neither 4 ps nor 20 ps runs had equilibrated at the temperatures nearest the transition (at 100 K the seed-averaged $|m|$ was still falling across the averaged half of both: from 0.49 in its first 0.5 ps to 0.38 in its last for the 4 ps run, and from 0.15 in its first 2 ps window to 0.07 in its last for the 20 ps run). Equilibration is checked explicitly for every reported point by comparing the mean of $|m|$ over consecutive 2 ps windows of the averaged half; the $\alpha = 0.05$ runs are kept alongside (`src/spin_dynamics/logs_v3_4ps`, `logs_v3_a005_20ps`) and approach the reported values from above.

For mp-1079285 the same machinery carries a three-shell model, $H = -J_b \sum_{\text{chain}} - J_{2b} \sum_{\pm 2b} - J_\perp \sum_{\text{inter}} \hat{s}_i \cdot \hat{s}_j + \sum_{\text{chain}} \mathbf{D} \cdot (\hat{s}_i \times \hat{s}_j)$ on the Fe sublattice (one spin per primitive cell, $|m| = 1.68\,\mu_B$ as a classical unit vector). Two `spin/exchange` instances carry the chain bond (cutoff 5 Å) and the 8.4 Å shell (Bethe–Slater radial form with its root placed at the chain distance); because the $\pm 2b$ bond has the same length as the inter-chain bonds, spins carry an atom type equal to their chain index modulo four, so that $\pm 2b$ pairs (types differing by two) receive J_{2b} by an explicit per-type override while inter-chain pairs (equal types) receive $J_\perp$. The DMI is carried by `spin/dmi` on the chain bond only (5 Å cutoff), with its vector calibrated – not assumed – by zero-step energy evaluations of the very 90° spirals used in the DFT, so that the model reproduces $E(+) - E(-)$ for both spiral planes. Before any sweep, zero-step evaluations of the $1 \times 2 \times 1$, $1 \times 4 \times 1$ $(++--)$, $2a$ and $2c$ cells reproduce the four DFT energy differences — the two chain cells exactly, the $2a$ and $2c$ cells to 0.5%, which is the residual of fitting one $J_\perp$ to two inequivalent shells — and the $++--$ check is what exposes a $\pm 2b$ bond silently given $J_\perp$. The production cell is an $8 \times 8 \times 8$ replication of the $1 \times 4 \times 1$ unit (2048 spins), started from FM along b, with the same Langevin protocol as above at fourteen temperatures from 5 to 300 K (`src/spin_dynamics/build_mp1079285_model.py`, `outputs/mp1079285_spin_model.json`).

III. RESULTS

A. DFT validation of the shortlisted FM population

Of the 26 candidates, all carrying an MP-recorded `ground_truth`, confidence-1.0 ferromagnetic label, 21 have their FM label overturned by a lower-energy antiferromagnetic or ferrimagnetic configuration (margins 0.19–24.68 meV/atom), 3 are confirmed as the true FM ground state (1.24–2.23 meV/atom below the next-lowest configuration), and 2 ($CaLaTaFeO_6$ with AFM 0.10 meV/atom below FM and $Ba_4Ta_{10}FeO_{30}$ at 0.00 meV/atom, both from tight pairs) sit within 0.10 meV/atom – twice the per-atom electronic-convergence tolerance (EDIFF = 5×10^{-5} eV/atom) of the loose protocol – and are read

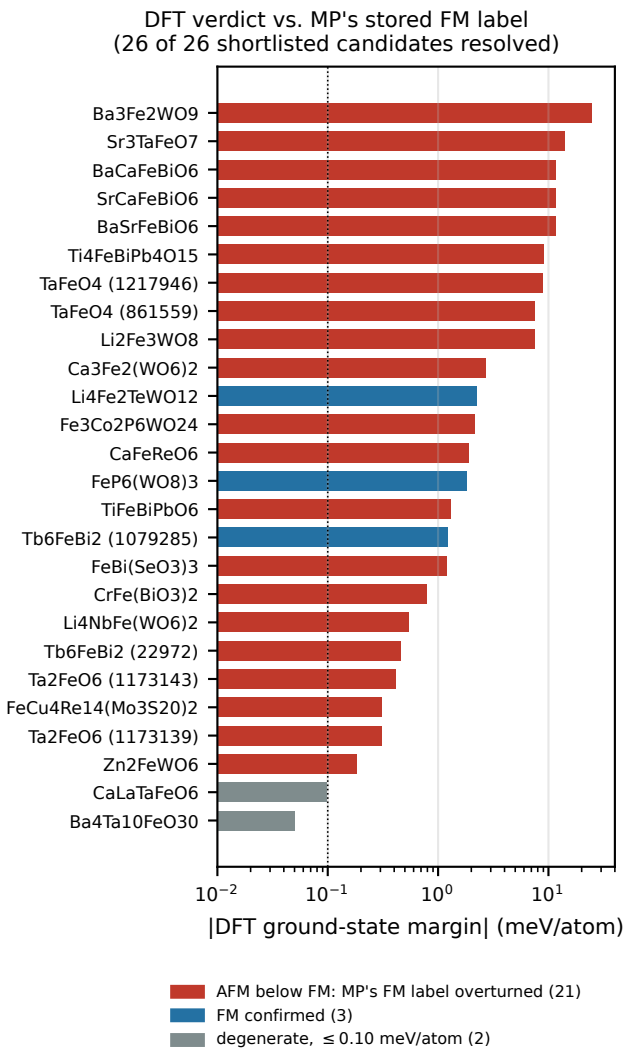


FIG. 4. DFT ground-state margin for the 26 candidates, sorted by magnitude (logarithmic horizontal axis; only calculations that converged and passed the spin-state audit enter, and a tight same-cell pair replaces the loose number wherever one exists). Twenty-one candidates' MP-recorded FM label is overturned by a lower-energy AFM or FiM configuration (red); three are confirmed as the true FM ground state (blue); two sit at or below the electronic-convergence tolerance ($\leq$ 0.10 meV/atom, gray) and are read as degenerate rather than informative. Duplicate formulas are distinguished by their MP identifier.

as degenerate rather than as confirming either ordering (Fig. 4, Table I). No candidate remains unresolved: the one configuration never completed (the re-diagnosed $1 \times 1 \times 2$ stacking of mp-682554, whose production run was dropped when the farm was stopped on 21 September to release the nodes) belongs to a candidate that another audited AFM cell already places 0.31 meV/atom below FM. The two shortlisted-but-labeled-FiM candidates (mp-22972, mp-551086) are counted relative to the FM alternative only, since a collinear FM/AFM enumeration cannot represent MP's recorded ferrimagnetic configuration directly; both have a collinear configuration below FM (an AFM cell 0.46 meV/atom below the FM twin of the same cell for mp-22972, a ferrimagnetic Fe/Cr cell 0.80 meV/atom below FM for mp-551086, both from tight pairs). That antiferromagnetic alignment wins this often in this particular set is unsurprising on chemical grounds: 23 of the 26 candidates are oxides, in which the magnetic cations are bridged by oxygen rather than being in direct contact – the geometry for which Goodenough–Kanamori–Anderson superexchange between half-filled d shells favours antiparallel alignment [67–69]. A workflow that initializes every relaxation ferromagnetically therefore starts each of these compounds from the chemically less likely state.

The tight pairs changed four verdicts, in both directions. Zn_2FeWO_6 (mp-1047414), degenerate at 0.10 meV/atom under the loose protocol, is AFM by 0.17–0.19 meV/atom in both of its same-cell pairs; $Ba_4Ta_{10}FeO_{30}$ (mp-1228547), "confirmed FM" by 1.18 meV/atom loosely, is exactly degenerate (0.001 meV/atom) once FM and AFM are compared in the same 90-atom cell; $TiFeBiPbO_6$ (mp-1216981), whose enumerated "AFM" cells had flipped only the d^0 Ti site, is AFM by 0.80–1.30 meV/atom in all four Fe-only supercells; and Tb_6FeBi_2 (mp-1079285) and $FeP_6(WO_8)_3$ (mp-850225) remain FM, with same-cell margins on the enumerated AFM configurations (3.25 and 3.38 meV/atom) larger than the loose numbers (2.32 and 2.09) suggested – although in both compounds an ordering computed later for the exchange model sets the margin: for mp-1079285 the inter-chain AFM orderings (Sec. III C) lie only 1.24–1.25 meV/atom above FM, and for mp-850225 a 68-atom cell with alternate Fe layers antiparallel along the hexagonal c axis lies 1.81 meV/atom above FM (Sec. IV C). For mp-850225, "FM" describes the Fe sublattice, which is the only part the enumeration varies. The W^{5+} moments (Sec. II D) align antiparallel to it, so the DFT ground state is strictly a collinear ferrimagnet, $2.0\,\mu_B$ per formula unit against $4.4\,\mu_B$ from Fe alone. All six loose-only overturns with $|\text{margin}| < 2$ meV/atom have been re-decided with tight pairs, and every one kept its sign: Ta_2FeO_6 (mp-1173143) $-0.41 \rightarrow -0.41$, $CrFe(BiO_3)_2$ (mp-551086) $-0.85 \rightarrow -0.80$, $FeBi(SeO_3)_3$ (mp-1197024) $-1.19 \rightarrow -1.19$, $CaFeReO_6$ (mp-1343817) $-1.77 \rightarrow -1.88$, the second Tb_6FeBi_2 entry (mp-22972) $-0.14 \rightarrow -0.46$, and $FeCu_4Re_{14}(Mo_3S_{20})_2$ (mp-682554) $-0.29 \rightarrow -0.31$ meV/atom. The last of these, a pair of 130-atom relaxations first cut off short of the force criterion when the farm was stopped, was later continued to convergence (29 and 32 ionic steps, final maximum forces 0.006 and 0.010 eV/Å); the gap moved from the 0.315 meV/atom of the cut-off runs to 0.313 meV/atom (40.7 meV per cell). A representative case, mp-861559 ($TaFeO_4$), illustrates the setup's internal consistency: its converged FM configuration reproduces MP's own recorded total magnetization to within $0.003\,\mu_B$ (30.00000 vs. $30.0022\,\mu_B$ for 36 atoms), ruling out a convention mismatch, while three symmetry-distinct AFM configurations all converge below it, the lowest by 0.274 eV/cell (≈45.7 meV/Fe).

Among the 24 candidates that resolve clearly (excluding the 2 degenerate ties), overturned outcomes outnumber confirmed FM outcomes 21 to 3. This sample is drawn from the top of the DMI-filtered shortlist (noncentrosymmetric, heavy-SOC-element, net-moment structures) rather than a random or stratified sample of the full `ground_truth` population, so it establishes the mislabeling rate within this specific structural family rather than a project-wide estimate; it is plausible, though untested here, that this family is more AFM-prone than the ground-truth population as a whole. What it establishes unambiguously is that MP's FM label cannot be taken at face value for candidates drawn from exactly this population, which is precisely the population that DMI/skyrmion-material screening pipelines – including the one used to generate this shortlist – select from by construction.

A provenance check against MP's own metadata (`theoretical` flag and ICSD [70] cross-references, queried 2026-09-20) adds a caveat that applies to the shortlist as a whole rather than to the ordering verdicts: 24 of the 26 staged candidates are flagged theoretical – hypothetical structures generated by MP's substitution and prototype workflows, 23 of them with no ICSD entry at all (mp-551086 carries an ICSD identifier but is still flagged theoretical). Only mp-1197024 ($FeBi(SeO_3)_3$, DFT ground state AFM) and mp-22972 (Tb_6FeBi_2, five ICSD entries) are experimentally reported compounds, and every candidate whose FM label survives DFT – including mp-754090 – is hypothetical, with MP hull distances of 0–0.085 eV/atom (mp-1079285 is a hypothetical polymorph of the experimental mp-22972). The funnel's symmetry and composition filters therefore preferentially select MP's generated structures, which is unsurprising – noncentrosymmetric cation orderings of otherwise centrosymmetric prototypes are exactly what substitution workflows produce – but it means a confirmed-FM verdict here is a statement about a candidate that may not be synthesizable, and any experimental follow-up should start from the two ICSD-backed entries (Fig. 5).

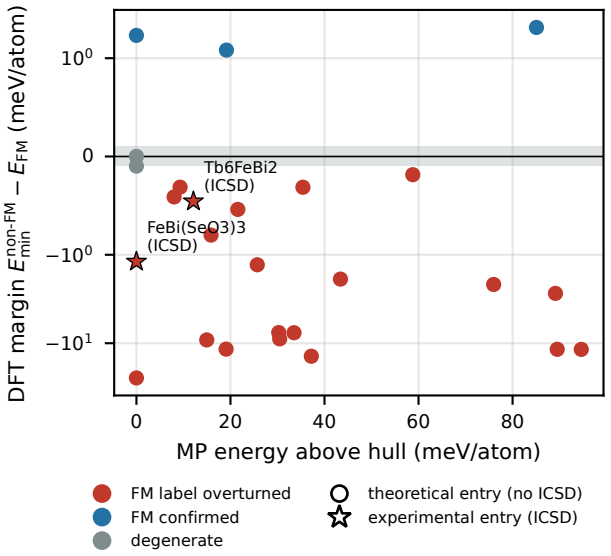


FIG. 5. DFT margin (lowest audited non-FM configuration minus FM, symmetric-log axis; negative means MP's FM label is overturned) against MP's own hull distance for the 26 candidates. Colour gives the verdict of Fig. 4; the gray band is the ±0.10 meV/atom degeneracy window. Only two candidates ($FeBi(SeO_3)_3$ and Tb_6FeBi_2 (mp-22972), stars) are experimentally reported ICSD compounds, both overturned; every confirmed-FM entry is a hypothetical structure, and the margin shows no trend with hull distance.

TABLE I. Summary of DFT-validated ordering outcomes for the 26 candidates. "Tight" counts the verdicts whose deciding margin comes from a same-cell FM/AFM pair (Sec. II D); the remaining verdicts rest on the loose MP-recipe comparison, one of them with a margin below 2 meV/atom (Sec. IV C).

Outcome	Count	Tight
MP FM label overturned (AFM/FiM)	21	9
FM confirmed	3	3
Degenerate ($\leq$ 0.10 meV/atom)	2	2
Total	26	14

B. mp-754090: DFT-derived exchange, DMI, and finite-temperature ordering

For mp-754090, confirmed FM by 2.23 meV/atom, the DFT-derived exchange and DMI constants are $J = 2.78$ meV and $D = 0.097$ meV per bond on the twelve-neighbor Fe sublattice, giving $D/J \approx 3.5\%$. This is notably smaller than an earlier order-of-magnitude placeholder estimate used before this calculation ($D/J \sim 25\%$), and is physically plausible: the magnetic ion at the probed bond, Fe, is a comparatively light 3d species with weak intrinsic spin-orbit coupling, despite the compound containing tungsten, a heavy-SOC element that is precisely what qualified this candidate for the DMI-compatibility funnel in the first place. This is a useful cautionary note for the screening heuristic itself: the presence of a heavy-SOC element anywhere in the composition does not guarantee a large DMI constant at the specific magnetic-ion bond that anchors the spin texture.

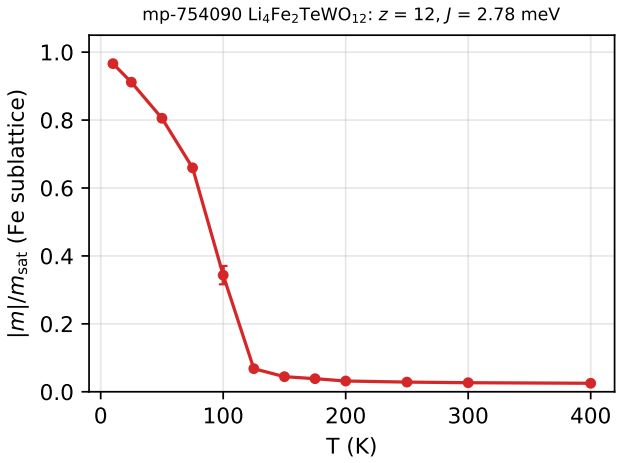


FIG. 6. Normalized net magnetization of the mp-754090 Fe sublattice (2048-spin cell) versus temperature, driven purely by the DFT-derived twelve-neighbor $J = 2.78$ meV with no fitting; 20 ps Langevin dynamics ($\alpha = 1$), mean over three thermostat seeds (error bars: seed-to-seed standard deviation, smaller than the symbols except at 100 K). Runs start from the ordered state, so the absolute scale is physical ($|m| \to 1$ as $T \to 0$); the high-temperature floor of ≈ 0.025 is the $1/\sqrt{N}$ finite-size value of a 2048-spin paramagnet.

Table II and Fig. 6 show the resulting finite-temperature magnetization sweep. $|m|$ decreases monotonically from 0.966 at 10 K through 0.805 at 50 K and 0.659 at 75 K to 0.343 at 100 K, collapses to 0.068 at 125 K, and then sits on the finite-size floor (0.025–0.044 for 150–400 K; the $1/\sqrt{N}$ value for $N = 2048$ is 0.022). The transition of the model therefore lies between 100 and 125 K for this cell size, with 100 K – the only point with an appreciable seed-to-seed spread (±0.027) and within-run fluctuation (0.043 standard deviation over the averaged half, and up to 0.049 between its 2 ps window means) – just below it; the finite-size-scaling analysis below sharpens this to a single temperature. Two zero-parameter estimates for the same Hamiltonian bracket this: the mean-field value $T_c^{MF} = zJ/3k_B = 129$ K ($z = 12$) and the classical-Heisenberg fcc value $k_B T_c \approx 3.18\,J$ [71], i.e. ≈ 103 K, which is the appropriate comparison for a twelve-coordinated close-packed classical unit-spin model. An independent Metropolis Monte Carlo calculation [72] of the same Hamiltonian on the same 2048-spin lattice (`src/spin_dynamics/mc_check_mp754090.py`; 2000 equilibration + 2000 measurement sweeps) gives $|m| = 0.967$, 0.912, 0.806, 0.661, 0.37 ± 0.03 and 0.07 ± 0.02 at 10, 25, 50, 75, 100 and 125 K, i.e. agrees with the Langevin sweep to within 0.002 below 100 K

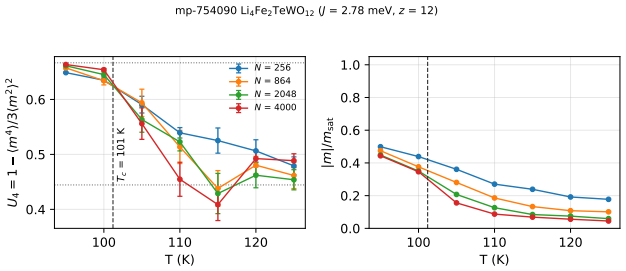


FIG. 7. Finite-size scaling of the mp-754090 spin model. Left: Binder cumulant $U_4 = 1 - \langle m^4\rangle/3\langle m^2\rangle^2$ at four system sizes with the box aspect ratio held fixed (N = 256, 864, 2048 and 4000 spins), from 40 ps Langevin dynamics ($\alpha = 1$) with the moments pooled over the last 20 ps of four thermostat seeds; error bars are a drop-one-seed jackknife. Dotted lines mark the ordered (2/3) and disordered (4/9) limits of a three-component order parameter and the dashed line the mean of the three pairwise crossings, T_c = 101 K. Right: $|m|(T)$ at the same four sizes, showing the collapse sharpening with N.

and within the seed spread at 100 K, so the reported curve is the equilibrium $|m|(T)$ of the model and not an artifact of the thermostat. The same Monte Carlo on the earlier six-neighbor lattice gives $|m|$(10 K) = 0.64 and 0.23 at 25 K – the no-long-range-order signature that exposed the cutoff error. The DMI control sweep (Sec. II G) changes $|m|$ by at most 0.013 (at 100 K, within the seed spread) and by ≤ 0.005 elsewhere, so an antisymmetric term of the measured 3.5% magnitude does not move the ordering scale of this model. The simulated curve is therefore quantitatively consistent with its own inputs; whether it is consistent with the real material depends on the single-J, zero-anisotropy, rigid-lattice approximations listed in Sec. IV C. For the record, the earlier versions of this sweep placed the knee at 150–200 K with the double-counted J = 12.06 meV and, once properly equilibrated, at 25–50 K with the six-neighbor (layered) model – both wrong for the reasons given in Sec. II G; the 8% loose/tight protocol spread in J shifts the transition by about 8 K, less than the temperature grid spacing.

A finite-size-scaling analysis pins this transition temperature down instead of bracketing it. Repeating the sweep at N = 256, 864, 2048 and 4000 spins with the box aspect ratio held fixed (so that every size sees the same twelve-neighbor shell; `src/spin_dynamics/run_fss.sh`) and forming the fourth-order Binder cumulant [73] $U_4 = 1 - \langle m^4\rangle/3\langle m^2\rangle^2$ of the three-component order parameter, which tends to 2/3 in the ordered phase and 4/9 in the disordered one and is size-independent at T_c, the $U_4(T)$ curves of consecutive sizes cross at 99.5, 101.3 and 102.7 K (Fig. 7, `outputs/mp754090_fss.csv`). The crossings drift upward monotonically with system size, the expected signature of corrections to scaling, giving $T_c = 101.2 \pm 1.3$ K over the three pairs with the largest pair ($N = 2048\times4000$, least affected by those corrections)

TABLE II. Normalized net magnetization $|m|$ of the mp-754090 Fe sublattice vs. temperature: 20 ps Langevin dynamics ($\alpha = 1$) per point from the ordered state, averaged over the last 10 ps and over three thermostat seeds ($\pm$ = seed-to-seed standard deviation), DFT-derived twelve-neighbor J = 2.78 meV, exchange only.

T (K)	$\lvert m\rvert$	T (K)	$\lvert m\rvert$
10	0.966 ± 0.000	150	0.044 ± 0.002
25	0.912 ± 0.001	175	0.039 ± 0.001
50	0.805 ± 0.002	200	0.032 ± 0.002
75	0.659 ± 0.004	250	0.029 ± 0.000
100	0.343 ± 0.027	300	0.027 ± 0.001
125	0.068 ± 0.001	400	0.025 ± 0.000

at 102.7 K. The susceptibility $\chi = N(\langle m^2\rangle - \langle|m|\rangle^2)/k_BT$ peaks at 105 K independently at all four sizes. The transition of this model is therefore the classical-Heisenberg fcc value ($\approx$ 103 K) to within the 5 K grid spacing, while mean field overestimates it by 28%. Two caveats attach to the number: U_4 is a ratio of small fluctuating moments, so its jackknife error grows from $\pm$0.0003–0.002 at 95 K to $\pm$0.03 above T_c, and in that noisy regime the difference between two sizes can change sign on scales far below its own error bar. The crossings quoted here are therefore the lowest-temperature sign changes that persist: the difference is required to stay positive at a majority of the higher temperatures measured, which discards both the sub-noise flips inside the ordered plateau, where every U_4 has converged on 2/3, and isolated single-point reversals above T_c, without asking any one noisy difference to carry the result (`src/spin_dynamics/analyze_fss.py`).

C. mp-1079285: exchange hierarchy and DMI

The same-cell pairs of Sec. II F give, for Tb_6FeBi_2, a strongly anisotropic exchange network. The chain pair (18-atom $1\times2\times1$ cell) places the chain-antiparallel state 58.43 meV per cell above FM, i.e. J_b = 14.61 meV for the 4.198 Å Fe–Fe bond. The two inter-chain pairs are nearly identical, ΔE_{2a} = 22.48 and ΔE_{2c} = 22.24 meV per 18-atom cell (1.25 and 1.24 meV/atom; Fe $\pm$1.64–1.67 μ_B, induced Tb $\leq$ 0.21 μ_B in all four calculations), which supports the $J_a \approx J_c$ part of the mean-field assumption and gives $J_\perp$ = 2.80 meV – a factor of 5.2 weaker than the chain coupling, but ferromagnetic, so the material is a three-dimensional ferromagnet of coupled FM chains rather than a quasi-one-dimensional one. The spiral pairs give $\mathbf{D}\cdot\hat{n}_1 = -0.074$ and $\mathbf{D}\cdot\hat{n}_2 = -0.063$ meV, $|\mathbf{D}| = 0.098$ meV per chain bond and $|\mathbf{D}|/J_b = 0.67\%$ – five times smaller than the D/J of mp-754090, again with Fe as the only magnetic ion and the heavy elements (Tb, Bi) contributing only through the ligand field. The $1\times4\times1$ $++--$ pair places that state 84.92 meV per 36-atom cell above FM; subtracting $4J_b$ from the chain pair leaves

$8J_{2b} = 26.50$ meV, i.e. $J_{2b} = 3.31$ meV – ferromagnetic and comparable to $J_\perp$, so the $\pm 2b$ bond is not a negligible correction: it carries 13% of $zJ = 2J_b + 2J_{2b} + 6J_\perp = 52.6$ meV and raises the mean-field estimate from $T_c^{\mathrm{MF}} = 178$ K (chain + inter-chain only) to 203 K. This value is a difference of two energies computed with different k-meshes ($3\times3\times3$ on the $1\times2\times1$ cell for J_b, $3\times2\times3$ on the $1\times4\times1$ cell for ΔE_{4b}). A same-cell check – the $+-+-$ chain pattern in the $1\times4\times1$ cell, which flips all four chain bonds ($\Delta E = 8J_b$) and is compared with the FM member of the $++--$ pair – gives $\Delta E = 100.45$ meV per 36-atom cell, i.e. $J_b = 12.56$ meV, 14% below the $1\times2\times1$ value (the relaxation, first cut off at ionic step 24, was continued to the force criterion: 28 ionic steps, final maximum force 0.007 eV/Å, lowering ΔE by 0.48 meV from the cut-off value). The two pairs differ in cell and mesh – the $1\times2\times1$ pair sits in the relaxed cell of the enumerated AFM configuration ($a = 8.372$, $c = 8.376$ Å) with a mesh equivalent to 6 points along b in the primitive cell, the $1\times4\times1$ pairs in the relaxed FM cell ($a = 8.407$, $c = 8.396$ Å) with 8 – so the spread is the cell-shape and k-convergence sensitivity of a metallic Fe chain, and it brackets the exchange model: with the same-cell J_b the residual is $J_{2b} = 4.34$ meV, $zJ = 50.6$ meV and $T_c^{\mathrm{MF}} = 196$ K, against $J_{2b} = 3.31$, $zJ = 52.6$ meV and 203 K with the chain-pair value; $|\mathbf{D}|/J_b$ ranges over 0.67–0.78%. The sweep below uses the chain-pair parameters, from which J_b, $|\mathbf{D}|/J_b$ and the finite-size-scaling model were derived; the same-cell model was swept at 10, 100, 125, 150 and 200 K to check whether the 4% change in zJ moves the ordering bracket — it does not: $|m| = 0.975$, 0.668, 0.497, 0.124 and 0.047 (three seeds each, window drift ≤ 0.008 except +0.05 at 150 K, the transition point, against a within-run fluctuation of ± 0.04 there) against 0.975, 0.672, 0.509, 0.128 and 0.048 for the chain-pair model at the same five temperatures, so the collapse sits between 125 and 150 K in both parametrizations.

Table III and Fig. 8 show the finite-temperature sweep of the three-shell model. $|m|$ falls smoothly from 0.988 at 5 K to 0.672 at 100 K and 0.509 at 125 K, then collapses to 0.128 at 150 K (the one point with a seed-to-seed spread, ± 0.012, and a within-run fluctuation of ± 0.05, as expected at the transition of a finite cell) and sits on the finite-size floor of 0.03–0.05 from 200 K. The ordering temperature of this cell size is therefore 125–150 K — the finite-size-scaling analysis below sharpens it to a single temperature — i.e. 0.6–0.75 T_c^{MF}, the usual reduction of the classical Heisenberg transition below mean field (for comparison the ratio is 0.72 for the simple-cubic lattice and 0.79 for the twelve-coordinated fcc one [71]; this Pm structure has $z = 2+2+6$ with unequal couplings, whose anisotropy is expected to push the ratio below both isotropic values). Every point passed the equilibration check: the seed-averaged change of $|m|$ between the first and last 2 ps windows of the averaged half is ≤ 0.008 at every temperature (and +0.018 at 150 K, against within-window fluctuations of ± 0.05 there), with no monotonic trend. The sweep is insensitive to the DMI

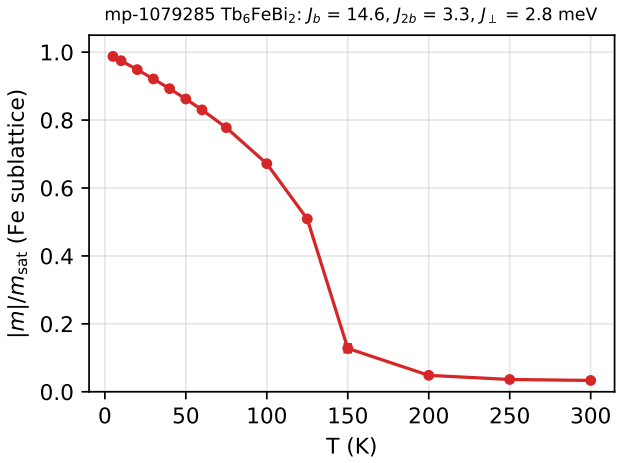


FIG. 8. Normalized net magnetization of the mp-1079285 Fe sublattice (2048-spin cell) versus temperature from the three-shell model with DFT-derived $J_b = 14.61$, $J_{2b} = 3.31$, $J_\perp = 2.80$ and $|\mathbf{D}| = 0.098$ meV; 20 ps Langevin dynamics ($\alpha = 1$) from the ordered state, mean over three thermostat seeds (error bars: seed-to-seed standard deviation, smaller than the symbols except at 150 K).

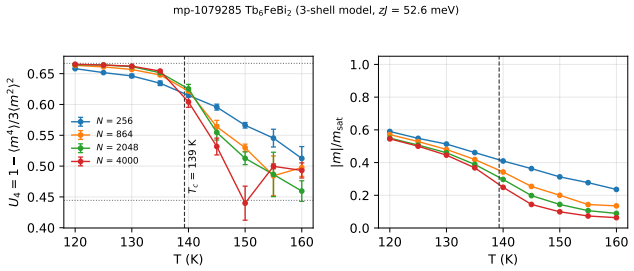


FIG. 9. Finite-size scaling of the mp-1079285 three-shell model, as Fig. 7: Binder cumulant (left) and $|m|$ (right) at $N = 256$, 864, 2048 and 4000 spins with the box aspect ratio held fixed. Four 40 ps seeds per point at 120–130 and 155–160 K; 10–14 seeds at 135–150 K, where the crossings lie, six of them 80 ps. The dashed line is the mean of the three pairwise crossings, $T_c = 139$ K.

at this D/J: $|\mathbf{D}|/J_b = 0.67\%$ produces a canting angle of order $D/2J \approx 0.2°$ per bond, far below the thermal disorder at any temperature on the grid.

The same finite-size-scaling analysis was run on this model (Fig. 9, `outputs/mp1079285_fss.csv`). The $U_4(T)$ curves of consecutive sizes cross at 140.9, 141.4 and 135.6 K, giving $T_c = 139.3 \pm 2.6$ K, and χ peaks at 145 K for the three smaller cells and 140 K for the largest. This falls inside the 125–150 K bracket read off the 2048-spin sweep and puts the transition at 0.68 T_c^{MF}, just below the isotropic simple-cubic ratio 0.72, as the exchange anisotropy leads one to expect. The crossings are not monotonic in system size as they are for mp-754090, and the analysis is harder here for two reasons. The exchange is strongly anisotropic — $J_b/J_\perp = 5.2$, though

ferromagnetic on every shell — so strong intra-chain correlations survive well above the three-dimensional ordering temperature and the U_4 curves separate gradually rather than pivoting about a point; and the fluctuations near T_c are correspondingly larger, with the jackknife error on U_4 growing from ±0.0002–0.0007 at 120 K to ±0.004–0.028 at 150 K. The 135–150 K window was therefore run at 80 ps with 10–14 seeds per point instead of 40 ps with four, which cut the error where the crossings lie (±0.048 → ±0.014 for $N = 4000$ at 145 K, though not uniformly: the same cell's error at 150 K grew from ±0.020 to ±0.028 as the longer runs sampled more of the disordered-phase excursions) and is what makes the three crossings agree to ±2.6 K; the 155 and 160 K points retain only four seeds each and scatter back above the disordered limit 4/9, and one such non-persistent sign change was discarded by the rule described above. The number to quote for this compound is therefore a transition near 140 K with a spread of a few kelvin, against 125–150 K from the single-cell sweep alone.

Both crossings were re-derived under changes to the sampling window to test how much of the quoted spread is statistical. The equilibration cut matters and cannot be relaxed: discarding only the first quarter of each run instead of the first half moves mp-754090 to 109.8 K and mp-1079285 to 134.5 K, because the retained data still contain the thermalization transient from the ordered start. With the half-discard in place, mp-754090 is insensitive to further truncation — averaging only the last quarter gives 98.0/102.4/103.0 K, mean 101.1 K, against the published 101.2 K. mp-1079285 is not: its last quarter gives 141.0/141.1/123.4 K, mean 135.2 K, and the entire shift comes from the 2048 × 4000 pair. A check that shortens no window — analyzing only the six 80 ps runs, which cover 135–150 K — returns 140.0/143.2/135.7 K, mean 139.6 K, within 0.3 K of the published value. The instability of individual pairs is visible directly in the raw series: one 40 ps seed at $N = 4000$, 140 K has $U_4 = 0.582$ in its third quarter and 0.383 in its fourth (a 10 ps window each), so the autocorrelation time of U_4 near T_c is comparable to the run length. Pooling 10–14 seeds suppresses the residual quarter-to-quarter difference of the pooled estimate to ≤ 0.039 but does not eliminate it. The ±1.3 and ±2.6 K spreads over pairwise crossings should therefore be read as lower bounds on the uncertainty of the model T_c, before the DFT coupling uncertainties of Sec. IV C are added.

TABLE III. Normalized net magnetization $|m|$ of the mp-1079285 Fe sublattice vs. temperature: 20 ps Langevin dynamics ($\alpha = 1$) per point from the ordered state, averaged over the last 10 ps and over three thermostat seeds (± = seed-to-seed standard deviation), DFT-derived three-shell model.

T (K)	$\lvert m\rvert$	T (K)	$\lvert m\rvert$
5	0.988 ± 0.000	60	0.830 ± 0.003
10	0.975 ± 0.000	75	0.778 ± 0.003
20	0.949 ± 0.001	100	0.672 ± 0.003
30	0.921 ± 0.001	125	0.509 ± 0.003
40	0.892 ± 0.002	150	0.128 ± 0.012
50	0.862 ± 0.002	200	0.048 ± 0.002
		250	0.036 ± 0.001
		300	0.033 ± 0.001

IV. DISCUSSION

A. Two independent routes to the same conclusion

Fahmy [15] detects the Materials Project's ferromagnetic bias statistically: a classifier trained on experimentally solved magnetic structures (MAGNDATA) disagrees with MP's stored label for thousands of compounds, at large N but without a first-principles check on any individual case. The present work detects the same qualitative phenomenon by the opposite route – a small-N, high-fidelity DFT energy-mapping check, applied specifically within the noncentrosymmetric, heavy-SOC-element, net-moment structural family that DMI/skyrmion-material discovery pipelines target. That these two very different methodologies, applied to different (though overlapping-in-spirit) populations, converge on the same conclusion – MP's single-initial-guess relaxation workflow systematically under-counts antiferromagnetic ground states – is stronger evidence than either result alone. It also sharpens the practical implication specifically for DMI-material discovery: the population this work checked is not a random cross-section of MP but exactly the slice that such pipelines filter down to, and a 21-of-26 overturn rate in that slice is a direct warning to any workflow, including our own, that uses MP's ordering label as a hard screening criterion rather than a prior.

B. Practical implication for DMI-candidate screening

The screening funnel used to generate this shortlist (Sec. II C) excludes AFM-labeled candidates from consideration entirely, on the reasoning that a net moment is required for the texture-hosting mechanisms this project targets. If 21 of 26 top-ranked, MP-labeled-FM candidates from that funnel are, in fact, AFM or ferrimagnetic, a nontrivial fraction of the true magnetic-material landscape relevant to this search is being filtered out before any DFT or simulation resource is spent on it – and, conversely, downstream compute (structure backfill, MLIP relaxation, spin-dynamics simulation) risks being spent on candidates whose net-moment premise is simply wrong. The corrective is not to trust the classifier over MP either – both are, in the end, estimates – but to treat any `ordering` field, ground-truth-labeled or classifier-predicted, as a prior to be checked with real DFT before committing expensive downstream stages, exactly as done here for mp-754090.

C. Limitations

The 26-candidate validation batch is drawn from the top of one DMI-filtered shortlist, not a random or stratified sample of the broader `ground_truth` population (1,864 candidates pass the DMI funnel, 1,195 with the hull as a hard filter; 26 of those, not 26 of the full database, were checked). The mislabeling rate reported here is a property of this structural family, not a project-wide estimate. The verdict rule is deliberately asymmetric: a candidate is called overturned as soon as one competing configuration lies below FM by more than 0.10 meV/atom, whereas "confirmed" requires every staged configuration to have passed the spin-state audit. The margins of the 12 overturned candidates decided on a loose (ISIF=3, relaxed-lattice) pair alone carry a protocol uncertainty that the sixteen doubly-computed configurations bound at 1.3 meV/atom. Only one of them has $|\text{margin}| < 2$ meV/atom ($Li_4NbFe(WO_6)_2$, -0.54), and it already has a tight pair on a second configuration that corroborates the sign (-0.47 meV/atom). The six sub-2 meV/atom overturns re-decided with tight pairs all kept their sign, with margins moving by at most 0.32 meV/atom (Sec. III A).

The mp-754090 DMI probe resolves one projection (the mean D_y of the eight inequivalent inter-sublattice bonds, in an arbitrary lab frame fixed consistently across the two canted calculations) of what is, in $P1$ symmetry, eight unconstrained three-component DMI vectors; a complete characterization would require at least eight independent canting probes, and without it the DMI cannot be placed in the spin model in a way that is calibrated against DFT (Sec. II G), so the mp-754090 sweep is exchange-only with the DMI effect bounded by a control. For mp-1079285 the symmetry is high enough (Pm) that two spiral pairs fix the full chain-bond vector, which is why its model does carry a calibrated DMI. The third confirmed ferromagnet, $FeP_6(WO_8)_3$ (mp-850225), has one Fe per rhombohedral cell, so its Fe sublattice is a Bravais lattice on which a two-sublattice canting probe sees no DMI at all (the $+\mathbf{t}$ and $-\mathbf{t}$ bonds contribute opposite $\mathbf{D} \cdot (\mathbf{S}_0 \times \mathbf{S}_1)$); resolving its D needs a spiral in the 102-atom hexagonal cell. That spiral pair (noncollinear with spin-orbit coupling, Fe directions constrained, W moments free) was run in three attempts (about 65 node-hours): one chirality converged (EDIFF $= 10^{-6}$ eV), but the other never reached self-consistency at the precision the extraction needs ($\Delta E = 18 \sin 120^\circ D_z \approx 15.6\, D_z$, i.e. about 1.6 meV for $D \sim 0.1$ meV), so no DMI is quoted for this compound. Its two exchange shells, six in-plane 8.77 Å and six out-of-plane 8.91 Å bonds, are separated by a second same-cell pair: with alternate hexagonal Fe layers antiparallel along c, every out-of-plane bond flips and no in-plane bond does, so $\Delta E = 12 J_2 = 123.2$ meV per 68-atom cell, giving $J_2 = 10.3$ meV and, with $8(J_1 + J_2) = 229.6$ meV from the first pair, $J_1 = 18.4$ meV, both ferromagnetic. The FM members of the two pairs (the same 68 atoms in different supercells) agree to 0.006 meV/atom. Because the W^{5+} moments are free to follow the Fe, these are effective Fe–Fe couplings that include the W polarization. Single-ion anisotropy was not probed in any compound and is set to zero in the spin dynamics, for lack of any DFT or literature basis for a value.

The magnetization sweeps are classical, unit-spin, rigid-lattice models with zero anisotropy. For mp-754090 the model is single-J: the in-plane and inter-layer bonds, and the four same-sublattice bonds, are all assigned the one J that a single FM/AFM pair determines; the Fe moment ($\approx 4\,\mu_B$ in DFT) is treated as a classical unit vector. For mp-1079285 the six inter-chain bonds share one $J_\perp$ by assumption. The Binder-cumulant crossings ($T_c = 101.2 \pm 1.3$ K for mp-754090, 139.3 ± 2.6 K for mp-1079285) are critical temperatures of the spin models, not of the materials: they inherit the uncertainty of the DFT-derived couplings (an 8% loose/tight spread in J for mp-754090, a 14% spread in J_b for mp-1079285) on top of the classical-unit-vector treatment of the moment, and the quoted error is only the spread over the three pairwise crossings. The quoted uncertainties are also unequally supported: the mp-754090 crossings are monotonic in system size, while those of the chain-anisotropic mp-1079285 model are not, and its 155–160 K points rest on four seeds each. Finally, the mp-754090 spin model was corrected twice during this work – a bond-double-counting error that inflated J by a factor of two, then a neighbor cutoff that silently discarded the six inter-layer bonds and left a lattice of uncoupled two-dimensional layers – and the Langevin sweeps had to be moved from $\alpha = 0.05$ to $\alpha = 1$ before they were equilibrated. Each of these was caught by a check that we now regard as mandatory: a zero-step energy evaluation against every DFT pair, an explicit count and geometry of the neighbor shells the cutoff admits, an independent Metropolis Monte Carlo calculation of $|m|(T)$ on the same lattice, and a window-drift test on every reported point.

V. CONCLUSION

Within the specific structural family that Dzyaloshinskii–Moriya-candidate screening pipelines select – noncentrosymmetric, net-moment, heavy-spin-orbit-coupling-element compounds – direct DFT energy-mapping on 26 Materials-Project-labeled ferromagnets finds 21 mislabeled (true ground state antiferromagnetic or ferrimagnetic), 3 confirmed, and 2 degenerate within 0.10 meV/atom. This DFT-level result independently corroborates, in a smaller but more directly consequential population, the large-scale statistical finding of systematic ferromagnetic bias in MP recently reported from ML classifiers trained on experimentally solved magnetic structures [15]. For one confirmed candidate, mp-754090, we additionally demonstrate a complete DFT-to-dynamics pipeline: the exchange constant extracted from first principles, with

no fitting, drives a finite-temperature spin-dynamics simulation (the DFT-resolved DMI projection, $D/J \approx 3.5\%$, is bounded by a control sweep rather than carried in the model) whose Binder-cumulant crossing over four system sizes places the transition at 101 ± 1 K, matching the classical-Heisenberg estimate for the same exchange constant and an independent Monte Carlo calculation; for a second confirmed candidate, mp-1079285, a three-shell model with a symmetry-resolved, DFT-calibrated chain DMI orders at 139 ± 3 K. Both results argue for treating high-throughput database ordering labels as a screening prior rather than ground truth, particularly within the structural families that downstream physical mechanisms (here, DMI) specifically select for.

DATA AVAILABILITY

The harmonized database schema, the classifier and screening code, the scripts that generate every VASP input from its Materials Project structure, the per-candidate verdict tables (`outputs/dft_verdicts.csv`, `outputs/dft_tight_validation.csv`), and the spin-model, spin-dynamics and finite-size-scaling scripts are available at https://github.com/akhtar07/magnetic-materials-dmi-screening.

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[1] I. Dzyaloshinsky, A thermodynamic theory of "weak" ferromagnetism of antiferromagnetics, J. Phys. Chem. Solids **4**, 241 (1958).
[2] T. Moriya, Anisotropic superexchange interaction and weak ferromagnetism, Phys. Rev. **120**, 91 (1960).
[3] A. Bogdanov and A. Hubert, Thermodynamically stable magnetic vortex states in magnetic crystals, J. Magn. Magn. Mater. **138**, 255 (1994).
[4] A. Fert, V. Cros, and J. Sampaio, Skyrmions on the track, Nat. Nanotechnol. **8**, 152 (2013).
[5] N. Nagaosa and Y. Tokura, Topological properties and dynamics of magnetic skyrmions, Nat. Nanotechnol. **8**, 899 (2013).
[6] S. Mühlbauer, B. Binz, F. Jonietz, C. Pfleiderer, A. Rosch, A. Neubauer, R. Georgii, and P. Böni, Skyrmion Lattice in a Chiral Magnet, Science **323**, 915 (2009).
[7] X. Z. Yu, Y. Onose, N. Kanazawa, J. H. Park, J. H. Han, Y. Matsui, N. Nagaosa, and Y. Tokura, Real-space observation of a two-dimensional skyrmion crystal, Nature **465**, 901 (2010).
[8] S. Heinze, K. von Bergmann, M. Menzel, J. Brede, A. Kubetzka, R. Wiesendanger, G. Bihlmayer, and S. Blügel, Spontaneous atomic-scale magnetic skyrmion lattice in two dimensions, Nat. Phys. **7**, 713 (2011).
[9] N. Romming, C. Hanneken, M. Menzel, J. E. Bickel, B. Wolter, K. von Bergmann, A. Kubetzka, and R. Wiesendanger, Writing and Deleting Single Magnetic Skyrmions, Science **341**, 636 (2013).
[10] S. Woo, K. Litzius, B. Krüger, *et al.*, Observation of room-temperature magnetic skyrmions and their current-driven dynamics in ultrathin metallic ferromagnets, Nat. Mater. **15**, 501 (2016).
[11] A. Fert, N. Reyren, and V. Cros, Magnetic skyrmions: advances in physics and potential applications, Nat. Rev. Mater. **2**, 17031 (2017).
[12] Y. Tokura and N. Kanazawa, Magnetic skyrmion materials, Chem. Rev. **121**, 2857 (2021).
[13] A. Jain, S. P. Ong, G. Hautier, W. Chen, W. D. Richards, S. Dacek, S. Cholia, D. Gunter, D. Skinner, G. Ceder, and K. A. Persson, Commentary: The Materials Project: A materials genome approach to accelerating materials innovation, APL Mater. **1**, 011002 (2013).
[14] M. K. Horton, J. H. Montoya, M. Liu, and K. A. Persson, High-throughput prediction of the ground-state collinear magnetic order of inorganic materials using Density Functional Theory, npj Comput. Mater. **5**, 64 (2019).
[15] A. E. Fahmy, Learning Magnetic Order Classification from Large-Scale Materials Databases, arXiv:2509.05909v2 (2026).
[16] S. V. Gallego, J. M. Perez-Mato, L. Elcoro, E. S. Tasci, R. M. Hanson, K. Momma, and M. I. Aroyo, MAGNDATA: towards a database of magnetic structures. I. The commensurate case, J. Appl. Crystallogr. **49**, 1750 (2016).
[17] S. V. Gallego, J. M. Perez-Mato, L. Elcoro, E. S. Tasci, R. M. Hanson, and G. Madariaga, MAGNDATA: towards a database of magnetic structures. II. The incommensurate case, J. Appl. Crystallogr. **49**, 1941 (2016).
[18] S. P. Ong, S. Cholia, A. Jain, M. Brafman, D. Gunter, G. Ceder, and K. A. Persson, The Materials Application Programming Interface (API): A simple, flexible and efficient API for materials data based on REpresentational State Transfer (REST) principles, Comput. Mater. Sci. **97**, 209 (2015).
[19] K. Choudhary, K. F. Garrity, A. C. E. Reid, *et al.*, The joint automated repository for various integrated simulations (JARVIS) for data-driven materials design, npj Comput. Mater. **6**, 173 (2020).
[20] S. Curtarolo, W. Setyawan, G. L. W. Hart, *et al.*, AFLOW: An automatic framework for high-throughput materials discovery, Comput. Mater. Sci. **58**, 218 (2012).
[21] J. Zhou, L. Shen, M. D. Costa, *et al.*, 2DMatPedia, an open computational database of two-dimensional materials from top-down and bottom-up approaches, Sci. Data **6**, 86 (2019).

[22] J. E. Saal, S. Kirklin, M. Aykol, B. Meredig, and C. Wolverton, Materials Design and Discovery with High-Throughput Density Functional Theory: The Open Quantum Materials Database (OQMD), JOM **65**, 1501 (2013).

[23] S. Kirklin, J. E. Saal, B. Meredig, A. Thompson, J. W. Doak, M. Aykol, S. Rühl, and C. Wolverton, The Open Quantum Materials Database (OQMD): assessing the accuracy of DFT formation energies, npj Comput. Mater. **1**, 15010 (2015).

[24] C. Draxl and M. Scheffler, NOMAD: The FAIR concept for big data-driven materials science, MRS Bull. **43**, 676 (2018).

[25] G. Ke, Q. Meng, T. Finley, T. Wang, W. Chen, W. Ma, Q. Ye, and T.-Y. Liu, LightGBM: A highly efficient gradient boosting decision tree, Adv. Neural Inf. Process. Syst. **30** (2017).

[26] L. Ward, A. Agrawal, A. Choudhary, and C. Wolverton, A general-purpose machine learning framework for predicting properties of inorganic materials, npj Comput. Mater. **2**, 16028 (2016).

[27] J. Schmidt, M. R. G. Marques, S. Botti, and M. A. L. Marques, Recent advances and applications of machine learning in solid-state materials science, npj Comput. Mater. **5**, 83 (2019).

[28] J. Nelson and S. Sanvito, Predicting the Curie temperature of ferromagnets using machine learning, Phys. Rev. Mater. **3**, 104405 (2019).

[29] C. J. Court and J. M. Cole, Magnetic and superconducting phase diagrams and transition temperatures predicted using text mining and machine learning, npj Comput. Mater. **6**, 18 (2020).

[30] N. C. Frey, M. K. Horton, J. M. Munro, S. M. Griffin, K. A. Persson, and V. B. Shenoy, High-throughput search for magnetic and topological order in transition metal oxides, Sci. Adv. **6**, eabd1076 (2020).

[31] A. Kabiraj, M. Kumar, and S. Mahapatra, High-throughput discovery of high Curie point two-dimensional ferromagnetic materials, npj Comput. Mater. **6**, 35 (2020).

[32] A. Fert and P. M. Levy, Role of anisotropic exchange interactions in determining the properties of spin-glasses, Phys. Rev. Lett. **44**, 1538 (1980).

[33] A. Crépieux and C. Lacroix, Dzyaloshinsky–Moriya interactions induced by symmetry breaking at a surface, J. Magn. Magn. Mater. **182**, 341 (1998).

[34] H. Yang, A. Thiaville, S. Rohart, A. Fert, and M. Chshiev, Anatomy of Dzyaloshinskii-Moriya interaction at Co/Pt interfaces, Phys. Rev. Lett. **115**, 267210 (2015).

[35] I. Batatia, D. P. Kovács, G. N. C. Simm, C. Ortner, and G. Csányi, MACE: Higher order equivariant message passing neural networks for fast and accurate force fields, Adv. Neural Inf. Process. Syst. **35** (2022).

[36] I. Batatia, P. Benner, Y. Chiang, *et al.*, A foundation model for atomistic materials chemistry, arXiv:2401.00096 (2023).

[37] E. Bitzek, P. Koskinen, F. Gähler, M. Moseler, and P. Gumbsch, Structural Relaxation Made Simple, Phys. Rev. Lett. **97**, 170201 (2006).

[38] A. Hjorth Larsen, J. Jørgen Mortensen, J. Blomqvist, *et al.*, The atomic simulation environment—a Python library for working with atoms, J. Phys.: Condens. Matter **29**, 273002 (2017).

[39] C. Chen and S. P. Ong, A universal graph deep learning interatomic potential for the periodic table, Nat. Comput. Sci. **2**, 718 (2022).

[40] B. Deng, P. Zhong, K. Jun, J. Riebesell, K. Han, C. J. Bartel, and G. Ceder, CHGNet as a pretrained universal neural network potential for charge-informed atomistic modelling, Nat. Mach. Intell. **5**, 1031 (2023).

[41] S. P. Ong, W. D. Richards, A. Jain, *et al.*, Python Materials Genomics (pymatgen): A robust, open-source python library for materials analysis, Comput. Mater. Sci. **68**, 314 (2013).

[42] J. P. Perdew, K. Burke, and M. Ernzerhof, Generalized Gradient Approximation Made Simple, Phys. Rev. Lett. **77**, 3865 (1996).

[43] S. L. Dudarev, G. A. Botton, S. Y. Savrasov, C. J. Humphreys, and A. P. Sutton, Electron-energy-loss spectra and the structural stability of nickel oxide: An LSDA+U study, Phys. Rev. B **57**, 1505 (1998).

[44] L. Wang, T. Maxisch, and G. Ceder, Oxidation energies of transition metal oxides within the GGA+U framework, Phys. Rev. B **73**, 195107 (2006).

[45] A. Jain, G. Hautier, S. P. Ong, C. J. Moore, C. C. Fischer, K. A. Persson, and G. Ceder, Formation enthalpies by mixing GGA and GGA+U calculations, Phys. Rev. B **84**, 045115 (2011).

[46] H. J. Monkhorst and J. D. Pack, Special points for Brillouin-zone integrations, Phys. Rev. B **13**, 5188 (1976).

[47] P. Hohenberg and W. Kohn, Inhomogeneous Electron Gas, Phys. Rev. **136**, B864 (1964).

[48] W. Kohn and L. J. Sham, Self-Consistent Equations Including Exchange and Correlation Effects, Phys. Rev. **140**, A1133 (1965).

[49] G. Kresse and J. Furthmüller, Efficient iterative schemes for ab initio total-energy calculations using a plane-wave basis set, Phys. Rev. B **54**, 11169 (1996).

[50] P. E. Blöchl, Projector augmented-wave method, Phys. Rev. B **50**, 17953 (1994).

[51] G. Kresse and D. Joubert, From ultrasoft pseudopotentials to the projector augmented-wave method, Phys. Rev. B **59**, 1758 (1999).

[52] P. E. Blöchl, O. Jepsen, and O. K. Andersen, Improved tetrahedron method for Brillouin-zone integrations, Phys. Rev. B **49**, 16223 (1994).

[53] A. I. Liechtenstein, M. I. Katsnelson, V. P. Antropov, and V. A. Gubanov, Local spin density functional approach to the theory of exchange interactions in ferromagnetic metals and alloys, J. Magn. Magn. Mater. **67**, 65 (1987).

[54] H. J. Xiang, E. J. Kan, S.-H. Wei, M. Whangbo, and X. G. Gong, Predicting the spin-lattice order of frustrated systems from first principles, Phys. Rev. B **84**, 224429 (2011).

[55] H. Xiang, C. Lee, H.-J. Koo, X. Gong, and M.-H. Whangbo, Magnetic properties and energy-mapping analysis, Dalton Trans. **42**, 823 (2013).

[56] M. Heide, G. Bihlmayer, and S. Blügel, Dzyaloshinskii-Moriya interaction accounting for the orientation of magnetic domains in ultrathin films: Fe/W(110), Phys. Rev. B **78**, 140403 (2008).

[57] F. Freimuth, S. Blügel, and Y. Mokrousov, Berry phase theory of Dzyaloshinskii–Moriya interaction and spin–orbit torques, J. Phys.: Condens. Matter **26**, 104202 (2014).

[58] D. Hobbs, G. Kresse, and J. Hafner, Fully unconstrained noncollinear magnetism within the projector augmented-wave method, Phys. Rev. B **62**, 11556 (2000).
[59] P.-W. Ma and S. L. Dudarev, Constrained density functional for noncollinear magnetism, Phys. Rev. B **91**, 054420 (2015).
[60] J. Tranchida, S. J. Plimpton, P. Thibaudeau, and A. P. Thompson, Massively parallel symplectic algorithm for coupled magnetic spin dynamics and molecular dynamics, J. Comput. Phys. **372**, 406 (2018).
[61] S. Plimpton, Fast Parallel Algorithms for Short-Range Molecular Dynamics, J. Comput. Phys. **117**, 1 (1995).
[62] A. P. Thompson, H. M. Aktulga, R. Berger, *et al.*, LAMMPS - a flexible simulation tool for particle-based materials modeling at the atomic, meso, and continuum scales, Comput. Phys. Commun. **271**, 108171 (2022).
[63] T. L. Gilbert, A phenomenological theory of damping in ferromagnetic materials, IEEE Trans. Magn. **40**, 3443 (2004).
[64] P.-W. Ma and S. L. Dudarev, Langevin spin dynamics, Phys. Rev. B **83**, 134418 (2011).
[65] B. Skubic, J. Hellsvik, L. Nordström, and O. Eriksson, A method for atomistic spin dynamics simulations: implementation and examples, J. Phys.: Condens. Matter **20**, 315203 (2008).
[66] R. F. L. Evans, W. J. Fan, P. Chureemart, T. A. Ostler, M. O. A. Ellis, and R. W. Chantrell, Atomistic spin model simulations of magnetic nanomaterials, J. Phys.: Condens. Matter **26**, 103202 (2014).
[67] P. W. Anderson, Antiferromagnetism. Theory of Superexchange Interaction, Phys. Rev. **79**, 350 (1950).
[68] J. B. Goodenough, Theory of the role of covalence in the perovskite-type manganites [La,M(II)]MnO_3, Phys. Rev. **100**, 564 (1955).
[69] J. Kanamori, Superexchange interaction and symmetry properties of electron orbitals, J. Phys. Chem. Solids **10**, 87 (1959).
[70] M. Hellenbrandt, The Inorganic Crystal Structure Database (ICSD)—Present and Future, Crystallogr. Rev. **10**, 17 (2004).
[71] K. Chen, A. M. Ferrenberg, and D. P. Landau, Static critical behavior of three-dimensional classical Heisenberg models: A high-resolution Monte Carlo study, Phys. Rev. B **48**, 3249 (1993).
[72] N. Metropolis, A. W. Rosenbluth, M. N. Rosenbluth, A. H. Teller, and E. Teller, Equation of State Calculations by Fast Computing Machines, J. Chem. Phys. **21**, 1087 (1953).
[73] K. Binder, Finite size scaling analysis of ising model block distribution functions, Z. Phys. B **43**, 119 (1981).